RLC PARALLEL CIRCUIT

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Content:

- Code overview
- Theoretical aspects
- Interface
- Bibliography

CODE OVERVIEW

MAIN.m

Editor - C:\Users\madel\OneDrive\Desktop\Rusu Ma

MAIN.m

```
clear all
close all
clc
R = 100;
C = 0.0001;
L = 1;
fr = 15;
E = 20;
N = 1;

menu(R, C, L, fr, E, N);
```

- *clear all* – clears all variables from workspace
- *close all* – closes all open figure windows
- *clc* – clears command window
- *rest of the code assignes values to the parameters of the RLC circuit*
- *last line of code calls the function "menu"*

menu.m

- *figure* – creates new figure window
- *uimenu* – creates menu bar titled "Information", that contains a submenu for the documentation (two formats), the bibliography and a close button.

menu.m *

```
function menu(R, C, L, fr, E, N)
clear all
close all
clc

% figure
figure('Name','RLC parallel circuit - Rusu Madalina',...
    'Units','normalized',...
    'Position',[0.2 0.2 0.6 0.6],...
    'Color', '#ffbae7','NumberTitle','off');

%uimenu
f=uimenu('Label','Information');
  submenuf = uimenu(f, 'Label', 'Documentation');
  uimenu(submenuf, 'Label', 'PDF documentation',...
      'Callback', 'open(''Rusu_Madalina_Documentation.pdf'')');
  uimenu(submenuf, 'Label', 'PowerPoint documentation',...
      'Callback', 'open(''Rusu_Madalina_Documentation.pptx'')');
  uimenu(f,'Label','Bibliography','Callback','open(''bibliography.docx'')');
  uimenu(f,'Label','Out','Callback','close',...
 'Separator','on','Accelerator','Q');
```

menu.m

```
%title
uicontrol('Style','text',...
'Units','normalized',...
'FontWeight','bold',...
'Position',[0.35 0.8 0.3 0.05],...
'foregroundcolor','w',...
'Backgroundcolor','#dc68b3',...
'String','RLC parallel circuits',...
'FontSize',20);

%push for circuit
uicontrol('Style', 'pushbutton', ...
   'String', 'Open Circuit', ...
   'Units', 'normalized', ...
   'FontSize',10, ...
   'Position', [0.4 0.6 0.2 0.05], ...
   'FontWeight','bold',...
   'foregroundcolor','w',...
   'Backgroundcolor','#dc68b3',...
   'Callback', 'parallel(R, C, L, fr, E, N);');
```

- uicontrol *('Style' , 'text'…)* – creates a text label that contains the title of the project
- *uicontrol ('Style' , 'pushbutton'…)* – creates a pushbutton titled "Open circuit" that directs the user to the RLC circuit through the callback of the function "parallel"

- *uicontrol ('Style' , 'pushbutton'…)* – creates a pushbutton titled "Open PDF Documentation" that directs the user to the documentation of the project
- *uicontrol ('Style' , 'pushbutton'…)* – creates a pushbutton that closes the program (through 'Callback','close')

menu.m

```
%push for documentation
 uicontrol('Style', 'pushbutton', ...
   'String', 'Open PDF Documentation', ...
   'Units', 'normalized', ...
   'FontSize',10, ...
   'Position', [0.4 0.5 0.2 0.05], ...
   'FontWeight','bold',...
   'foregroundcolor','w',...
   'Backgroundcolor','#dc68b3',...
   'Callback', 'open(''Bibliography.pdf'')');

%close
uicontrol('Style', 'pushbutton', ...
   'String','Close menu', ...
   'Units','normalized', ...
   'FontSize',10, ...
   'FontWeight','bold',...
   'Position', [0.4 0.4 0.2 0.05], ...
   'foregroundcolor','w',...
   'Backgroundcolor','#dc68b3',...
   'Callback','close');
```

menu.m

```
 %text
 str={'Student: Rusu Maria Mădălina','Group: 2022','ETTI - UTCN'};
uicontrol('Style','text',...
'Units','normalized',...
'FontWeight','bold',...
'FontSize',10,...
'Position',[0.4 0.25 0.2 0.08],...
'foregroundcolor','w',...
'Backgroundcolor','#dc68b3',...
'String',str,...
'HorizontalAlignment','center');

%music
 audioFile = 'musica.mp3';
    if exist(audioFile, 'file') ~= 2
        disp('Audio file not found!');
        return;
    end
    persistent player;
    if isempty(player)
        [y, Fs] = audioread(audioFile);
        player = audioplayer(y, Fs);
    end
```

- uicontrol (*'Style'* , *'text'*…) – creates a text label that contains data about the student
- *check audio file existence*
- *program uses a persistent variable to ensure the audio is only loaded and the audio player object is only created once*

- two pushbuttons titled “Music, please!” and “ Stop music” that either start or stop the music through callbacks to two different functions (@playmusic / @stopmusic)

menu.m

```
start = uicontrol('Style', 'pushbutton',...
    'String', 'Music, please!', ...
    'Units', 'normalized',...
    'FontSize', 10,...
    'FontWeight', 'bold', ...
    'Position', [0.8 0.2 0.1 0.05],...
    'ForegroundColor', 'w', ...
    'BackgroundColor', '#dc68b3',...
    'Callback', @playmusic);
stopp = uicontrol('Style', 'pushbutton',...
    'String', 'Stop Music', ...
    'Units', 'normalized',...
    'FontSize', 10,...
    'FontWeight', 'bold', ...
    'Position', [0.8 0.15 0.1 0.05],...
    'ForegroundColor', 'w', ...
    'BackgroundColor', '#dc68b3',...
    'Callback', @stopmusic);
```

menu.m

```
        function playmusic(~, ~)
            if ~isplaying(player)
                disp('Let`s party !');
                play(player);
            else
                disp('Music is already playing.');
            end
        end
        function stopmusic(~, ~)
            if isplaying(player)
                disp('You stopped the party :(');
                stop(player);
            else
                disp('No music is currently playing.');
            end
        end
    end
```

- *function playmusic* – plays music and displays a message (depending if the audio file is already playing or not)
- *Function stopmusic* – stops music and displays a message (depending if the audio file is already playing or not)

parallel.m

- *figure* – creates new figure window
- *uimenu* – same menu bar as before, so it can be accessed from both figures
- uicontrol *('Style' , 'text'…)* – creates a text label that contains the title of the project

parallel.m *

```
function parallel(R, C, L, fr, E, N)
close all
clc

%figure
Fig = figure('Name','Representation','Units','normalized',...
    'Position',[0.2 0.2 0.6 0.6],'Color', '#ffbae7',...
    'NumberTitle','off');

%uimenu
f=uimenu('Label','Information');
  submenuf = uimenu(f, 'Label', 'Documentation');
  uimenu(submenuf, 'Label', 'PDF documentation',...
      'Callback', 'open(''Rusu_Madalina_Documentation.pdf'')');
  uimenu(submenuf, 'Label', 'PowerPoint documentation',...
      'Callback', 'open(''Rusu_Madalina_Documentation.pptx'')');
  uimenu(f,'Label','Bibliography','Callback','open(''bibliography.docx'')');
  uimenu(f,'Label','Out','Callback','close',...
 'Separator','on','Accelerator','Q');

%title
uicontrol('Style','text','Units','normalized','FontWeight','bold',...
'Position',[0.42 0.9 0.2 0.05],'foregroundcolor','w',...
'Backgroundcolor','#dc68b3',...
'String','RLC parallel circuit','FontSize',15);
```

parallel.m

```
%show box R,L,C,E
BOX = uibuttongroup('Visible','off',...
'ForegroundColor','black',...
'TitlePosition','centertop',...
'Tag','radiobutton',...
'Backgroundcolor','#dc68b3',...
'Position',[0.7 0.2 0.25 0.6]);

%value of R
uicontrol('Style','text','Units','normalized',...
'FontWeight','bold','Position',[0.15 0.8 0.1 0.1],...
'foregroundcolor','w','Backgroundcolor','#dc68b3',...
'String','R:','FontSize',20,'Parent',BOX);

%value of L
uicontrol('Style','text','Units','normalized',...
'FontWeight','bold','Position',[0.15 0.6 0.1 0.1],...
'foregroundcolor','w','Backgroundcolor','#dc68b3',...
'String','L:','FontSize',20,'Parent',BOX);

%value of C
uicontrol('Style','text','Units','normalized',...
'FontWeight','bold','Position',[0.15 0.4 0.1 0.1],...
'foregroundcolor','w','Backgroundcolor','#dc68b3',...
'String','C:','FontSize',20,'Parent',BOX);

%value of E
uicontrol('Style','text','Units','normalized',...
'FontWeight','bold','Position',[0.15 0.2 0.1 0.1],...
'foregroundcolor','w','Backgroundcolor','#dc68b3',...
'String','E:','FontSize',20,'Parent',BOX);
```

- *uibuttongroup* – fuction used to group related components, in this case text fields, under the parent name "BOX"
- *Next, for each of the elements of the group, R, L, C and E, we defined text fields that cannot be modified and we assigned them to the group using: 'Parent' , 'BOX'.*

- *uibuttongroup* – fuction used to group related components, in this case only one text field for the soul purpose of having a good page layout, under the parent name "BOX2"
- *Text field freq (for frequency) has been defined and assigned to the group using: 'Parent' , 'BOX2'.*

parallel.m

```
%show box for frequency
BOX2 = uibuttongroup('Visible','off',...
'ForegroundColor','black',...
'TitlePosition','centertop',...
'Tag','radiobutton',...
'Backgroundcolor','#dc68b3',...
'Position',[0.39 0.6 0.26 0.1]);

freq = uicontrol('Style','text',...
'Visible','off',...
'Units','normalized',...
'FontWeight','bold',...
'Position',[0.4 0.62 0.08 0.06],...
'foregroundcolor','w',...
'Backgroundcolor','#dc68b3',...
'String','freq:',...
'FontSize',20);
```

parallel.m *

```
%EDIT BUTTONS
%edit R
uicontrol('Style','edit','Units','normalized',...
'Position',[0.3 0.8 0.6 0.1],'foregroundcolor','w',...
'Backgroundcolor','#c04194','String',R,...
'FontSize',15,'Parent',BOX,...
'Callback','R = str2num(get(gco,''String'')); showgraph(N, R, L, C, fr, E, selection)');

%edit L
uicontrol('Style','edit','Units','normalized',...
'Position',[0.3 0.6 0.6 0.1],'foregroundcolor','w',...
'Backgroundcolor','#c04194','String',L,...
'FontSize',15,'Parent',BOX,...
'Callback','L = str2num(get(gco,''String'')); showgraph(N, R, L, C, fr, E, selection)');

%edit C
uicontrol('Style','edit','Units','normalized',...
'Position',[0.3 0.4 0.6 0.1],'foregroundcolor','w',...
'Backgroundcolor','#c04194','String',C,...
'FontSize',15,'Parent',BOX,...
'Callback','C = str2num(get(gco,''String'')); showgraph(N, R, L, C, fr, E, selection)');

%edit E
uicontrol('Style','edit',...
'Units','normalized',...
```

- *uicontrol ('Style' , 'edit'…) –* allows us to create editable text field for each element of both groups (R,L,C,E,f).
- *The callback: input is converted into a number and then stored into a variable. Then, the function "showgraph" is called which helps with visualizing the graph with the new values.*

parallel.m

```
%Slider N
Slider_N=uicontrol('Style','slide',...
    'Units','normalized',...
    'Visible','off',...
    'Position',[0.42 0.75 0.2 0.04],...
    'Min',0,'Max',10,'Value',N,...
    'SliderStep',[1/10 0.1],...
    'Backgroundcolor','#dc68b3',...
    'Callback','N = round((get(gco,''Value''))), showgraph(N, R, L, C, fr, E, selection)');

nameN=uicontrol('Style','text',...
'Units','normalized',...
'Visible','off',...
'Position',[0.47 0.8 0.1 0.03],...
'foregroundcolor','white',...
'Backgroundcolor','#dc68b3',...
'FontWeight','bold',...
'String','N periods');
```

- *uicontrol ('Style' , 'slide'…)* – creates a slider between a minimum value (0) and a maximum value(10), with a step of 1. *Callback: the retrieved value is rounded to the next integer on the slider. The result can be visualized on the graph by calling the function "showgraph".*
- *Then we create a text field for the slider.*

parallel.m

```
%Label for min/max
minN=uicontrol('Style','text',...
'Units','normalized',...
'Visible','off',...
'Position',[0.63 0.75 0.02 0.02],...
'Backgroundcolor','#dc68b3',...
'FontWeight','bold',...
'foregroundcolor','w',...
'String',num2str(get(Slider_N,'Max')));

maxN=uicontrol('Style','text',...
'Units','normalized',...
'Visible','off',...
'Position',[0.39 0.75 0.02 0.02],...
'Backgroundcolor','#dc68b3',...
'FontWeight','bold',...
'foregroundcolor','w',...
'String',num2str(get(Slider_N,'Min')));
```

- *Labels for min/max of the slider are defined as text fields. They have been assigned the values of 0 and 10, and we need to convert these numerical values into string so that they can pe displayed on the interface.*

parallel.m *

```
%popup menu for type of graph
selection = 1;
uicontrol('Style', 'popup',...
        'String', {'Impedance R', 'Impedance L', 'Impedance C', 'Total impedance',...
        'Equivalent impedance + Inductive reactance + Capacitive reactance',...
        'Currents it=iR+iL+iC', 'Voltages u=uR+uC+uR'},...
        'Units', 'normalized',...
        'Position', [0.08 0.5 0.2 0.05],...
        'BackgroundColor', '#dc68b3',...
        'ForegroundColor', 'white',...
        'Value',selection,...
        'Callback', 'selection = get(gco, ''Value''); showgraph(N, R, L, C, fr, E, selection)');
```

- *The popup menu is created using a variable called "selection" that is assigned a value based on the index selected by the user. By calling the function "showgraph", the correct graph is being displayed based on the value of the index.*

parallel.m

```
%radiobutton to show pic
radiobutton = 0;
uicontrol('Style','radiobutton',...
 'Units','normalized',...
 'BackgroundColor',[0.75,0.75,0.75],...
'ForegroundColor','k',...
 'Position',[0.1 0.85 0.11 0.05],...
 'FontWeight', 'bold', ...
 'BackgroundColor', '#dc68b3', ...
  'ForegroundColor', 'white', ...
 'FontAngle', 'italic', ...
 'Value',radiobutton,...
'String','Show circuit', ...
'Callback', 'radiobutton = get(gco, ''Value''); showpic(radiobutton)');

%checkbox for edit box
edits = uicontrol('Style', 'checkbox', ...
    'Units', 'normalized', ...
    'Position', [0.75 0.85 0.11 0.05], ...
    'String', 'Show edit box', ...
    'FontWeight', 'bold', ...
    'BackgroundColor', '#dc68b3', ...
    'ForegroundColor', 'white', ...
     'Callback', @show);
```

- *uicontrol ('Style' , 'radiobutton'…)* – creates a radiobutton used to display the electrical chematic of the circuit, through callback of the function "showpic"
- *uicontrol ('Style' , 'checkbox'…)* – creates a checkbox and, if it is checked, the two edit boxes previously mentioned appear on the interface

- *Here we created two pushbuttons: one that takes the user back to the main interface (or menu) through callback of the function "menu" and one that closes the interface completely.*

parallel.m

```
%go back
uicontrol('Style', 'pushbutton', ...
    'String','Back', ...
    'Units','normalized', ...
    'FontSize',10, ...
    'FontWeight','bold',...
    'Position', [0.9 0.1 0.06 0.05], ...
    'foregroundcolor','w',...
    'Backgroundcolor','#dc68b3',...
    'Callback', 'menu(R, C, L, fr, E, N);');

%close
uicontrol('Style', 'pushbutton', ...
    'String','Close', ...
    'Units','normalized', ...
    'FontSize',10, ...
    'FontWeight','bold',...
    'Position', [0.9 0.05 0.06 0.05], ...
    'foregroundcolor','w',...
    'Backgroundcolor','#dc68b3',...
    'Callback','close');
```

parallel.m

```
% fuction used to show/hide boxes and slider
function show(~, ~)
    if get(edits, 'Value') == 1
        set(BOX, 'Visible', 'on');
        set(BOX2, 'Visible', 'on');
        set(Slider_N, 'Visible', 'on')
        set(freq, 'Visible', 'on');
        set(nameN, 'Visible', 'on')
        set(maxN, 'Visible', 'on')
        set(minN, 'Visible', 'on');
    else
        set(BOX, 'Visible', 'off');
        set(BOX2, 'Visible', 'off');
        set(Slider_N, 'Visible', 'off')
        set(freq, 'Visible', 'off');
        set(nameN, 'Visible', 'off');
        set(maxN, 'Visible', 'off')
        set(minN, 'Visible', 'off');
    end
end

end
```

This function, called "show", allows the program to either display or not the two text and edit boxes and the slider (with name and labels) depending weather or not the checkbox is checked or not (has the value 0 => false, 1 =>true).

showpic.m

showpic.m

```
function showpic(radiobutton)

if radiobutton == 1
    img1 = imread('circuitrlc.png');
    ax1 = axes('units', 'normalized', 'position', [0.06 0.57 0.26 0.27]);
    image(ax1, img1);
    axis(ax1, 'off');
else
    if radiobutton == 0
    img1 = imread('background.png');
    ax1 = axes('units','normalized','position',[0.06 0.57 0.26 0.27]);
    image(ax1, img1);
    axis(ax1, 'off');
    end

end
```

This function with the parameter "radiobutton" displays a picture of the circuit, if the value of the parameter is 1, or a picture of the background, for value 0. The function reads the image from the folder and then it is displayed on a certain position.

THEORETICAL ASPECTS

(used in showgraph.m)

Definitions

- RLC circuits are electrical circuits consisting of a resistor (R), an inductor (L), and a capacitor (C), connected in series or in parallel. In this project, we consider a circuit where the components are connected in parallel.
- RLC circuits have many applications as oscillator circuits.
- An RLC circuit can be used as a band-pass filter, band-stop filter, low-pass filter or high-pass filter.

Formulas

Impedance R: Xr = R (first graph)

Impedance L: XL = 2 * pi * f * L (second graph)

Impedance C: XC = 1 ./ 2 * pi * f * C (third graph)

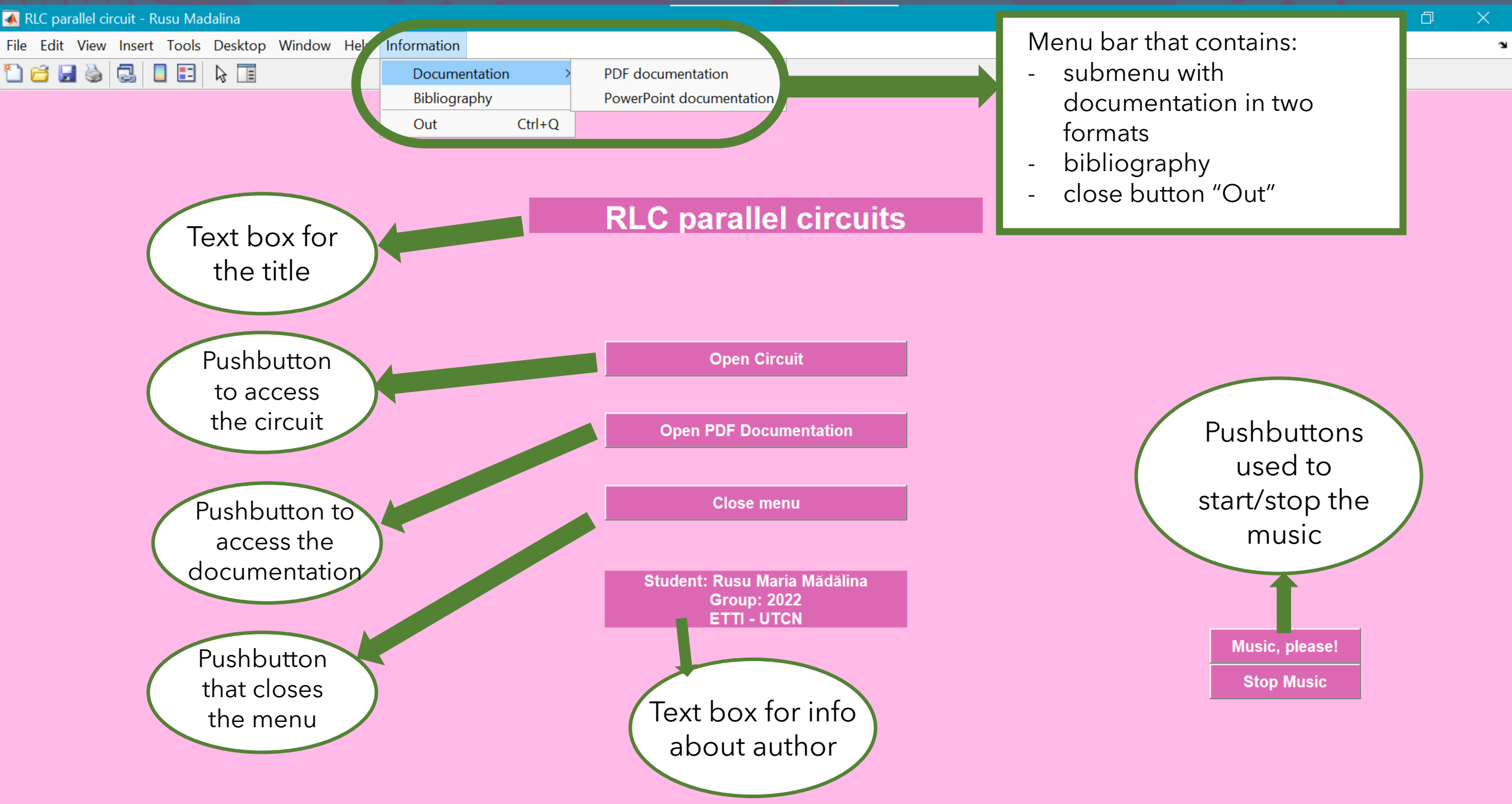
Total impedance of an RLC parallel circuit:

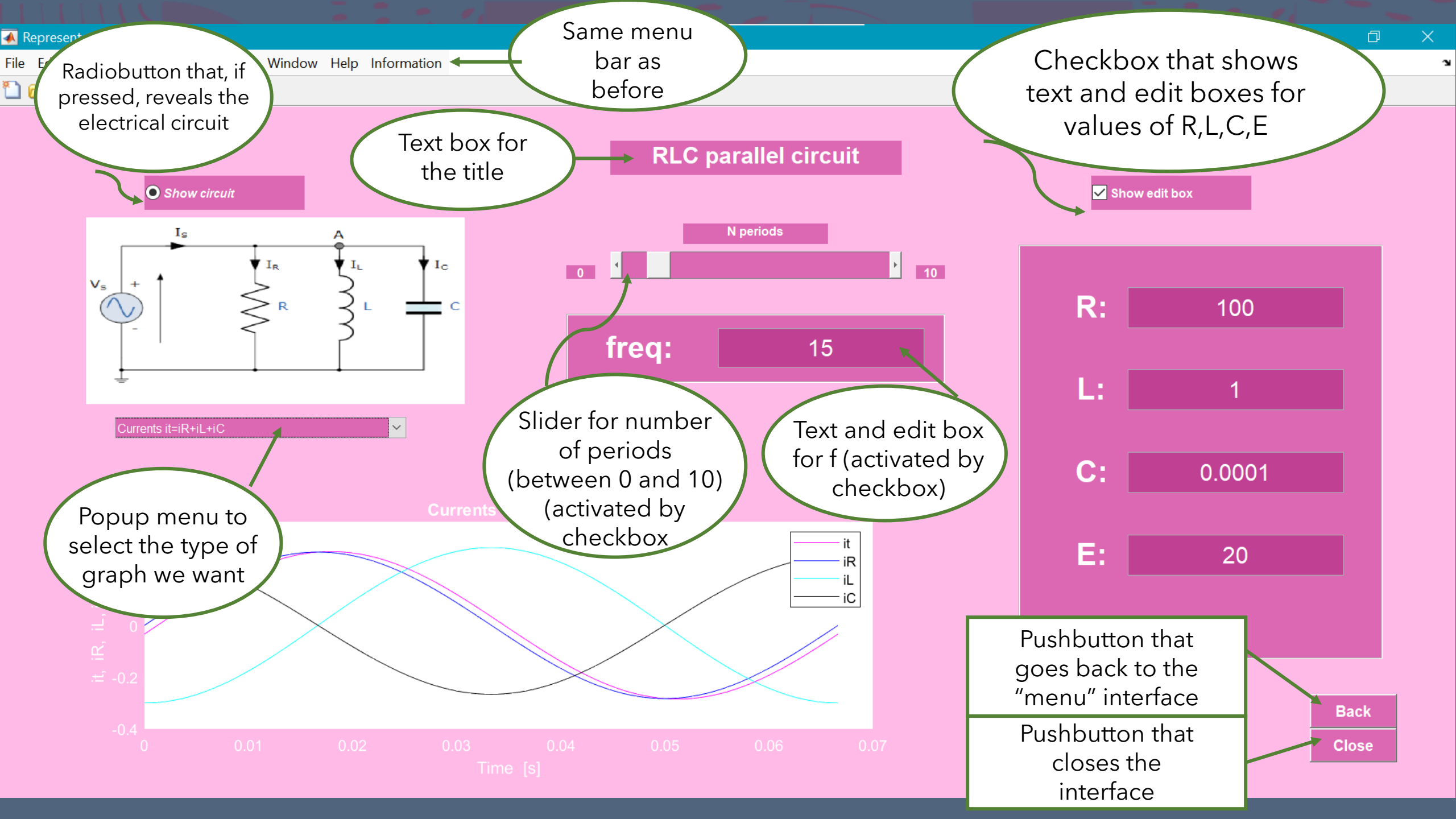
$$Z = \frac{1}{\sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_L} - \frac{1}{X_C}\right)^2}}$$

Equivalent impedance of an RLC parallel circuit:

$$Z_{AB} = Z_{Pech} = R \parallel Z_L \parallel Z_C = R \parallel \left(\frac{j\omega L}{1-\omega^2 LC}\right) = \frac{j\omega L}{1-\omega^2 LC + j\omega \frac{L}{R}}$$

INTERFACE


RLC parallel circuit - Rusu Madalina
File Edit View Insert Tools Desktop Window Help
Information
Documentation
Bibliography
Out Ctrl+Q
PDF documentation
PowerPoint documentation
Menu bar that contains:
- submenu with documentation in two formats
- bibliography
- close button "Out"
RLC parallel circuits
Text box for the title
Open Circuit
Pushbutton to access the circuit
Open PDF Documentation
Pushbutton to access the documentation
Close menu
Pushbutton that closes the menu
Student: Rusu Maria Mădălina
Group: 2022
ETTI - UTCN
Text box for info about author
Pushbuttons used to start/stop the music
Music, please!
Stop Music


Represent
File
Window
Help
Information
Same menu bar as before
Radiobutton that, if pressed, reveals the electrical circuit
Text box for the title
RLC parallel circuit
Checkbox that shows text and edit boxes for values of R,L,C,E
Show circuit
Show edit box
N periods
0
10
freq:
15
R:
100
L:
1
C:
0.0001
E:
20
Currents it=iR+iL+iC
Slider for number of periods (between 0 and 10) (activated by checkbox
Text and edit box for f (activated by checkbox)
Popup menu to select the type of graph we want
Currents
it
iR
iL
iC
it, iR, iL,
0
-0.2
-0.4
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Time [s]
Pushbutton that goes back to the "menu" interface
Pushbutton that closes the interface
Back
Close

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THANK YOU!